10 Heron's Formula

A triangular sail design makes a boat possible to travel against the wind, using a technique known as tacking. To make a triangular sail, Heron's formula is used to calculate the area of the required cloth to make by using the measurements of the sail that have to form.

What's Inside

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TOPIC 1

AREA OF A TRIANGLE

Learning Objectives

➥ Students will be learning the different types of triangles.

Learning Outcomes

➥ Students will be able to apply different formulas related to the area and perimeter of triangles.

Real Life Application

In our everyday lives, we frequently see triangles at sign boards. These signs are shaped like an equilateral triangle, suggesting that their three sides are equal in length and their angles are also equal. They all are made of equal size by using area measurement.

These sign boards are helpful in indicating the signs which are necessary for a particular situation. For example, if somewhere construction is going on, then they place a signboard which tells you to stop and not to move further. This is how all the signboards are made up of using the concept of the area of a triangle.

This represents a basic real life application of the area of triangles.

In this topic, we will be studying the area of triangles in detail.

A plane figure, bounded by three line segments, is called a triangle.

We know that Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

When the triangle is right-angled, we can directly apply the formula by using two sides containing the height and the right angle at its base.

Let us find out the area of the equilateral and isosceles triangles using the above formula.

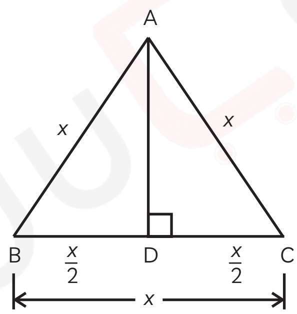
Equilateral Triangle

An equilateral triangle, also known as a regular triangle, is a triangle with all three sides of equal length.

Let $AB = BC = AC = x$

Also, $\angle A = \angle B = \angle C = 60^o$

Now, draw a perpendicular from vertex A, which divides BC into two equal parts, *i.e.*,



$$BD = DC = \frac{BC}{2}$$

Hence, AD is the height of the equilateral triangle ABC.

From the figure, in ΔABD,

$$AB^2 = AD^2 + BD^2 \qquad \text{[By Pythagoras' theorem]}$$

$$x^2 = AD^2 + \left(\frac{x}{2}\right)^2$$

$$AD^2 = x^2 - \left(\frac{x}{2}\right)^2$$

$$AD^2 = x^2 - \frac{x^2}{4}$$

$$AD^2 = \frac{4x^2 - x^2}{4}$$

$$AD^2 = \frac{3x^2}{4}$$

$$AD = \frac{\sqrt{3}x}{2}$$

Now,

Area of triangle ABC= $\frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times BC \times AD$$

$$= \frac{1}{2} \times x \times \frac{\sqrt{3}x}{2}$$

Area of equilateral triangle= $\frac{\sqrt{3}}{4}x^2 = \frac{\sqrt{3}}{4} \times (\text{Side})^2$

Important

➥ *In an equilateral triangle, each angle is equal to 60° and all angles are congruent.*

➥ *The equilateral triangle is divided into equal halves by a perpendicular, running from the vertex to the opposite side.*

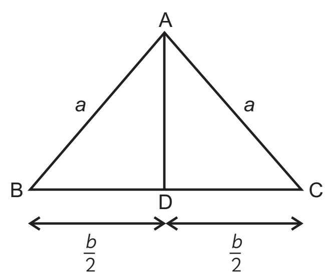
➥ *The perimeter of an equilateral triangle is 3 × Side = 3x*

➥ *Altitude/Height =* $\frac{\sqrt{3}}{2}x$

Isosceles Triangle

An isosceles triangle is a triangle that has any two sides equal in length. The two angles of an isosceles triangle, opposite to equal sides, are equal in measure.

Let, ABC be an isosceles triangle, such that AB = AC = a and BC = b



Draw a perpendicular from vertex A which divides BC into two equal parts, *i.e.*,

$\therefore$ $BD = DC = \frac{BC}{2}$

AD represents the height of the ABC and also divides it into two congruent right-angled triangles ADB and ADC.

We can find AD using Pythagoras' theorem in anyone of the right-angled triangles.

As $\quad AB^2 = AD^2 + BD^2$

$$a^2 = AD^2 + \left(\frac{b}{2}\right)^2$$

$$AD^2 = a^2 - \frac{b^2}{4}$$

$$AD = \sqrt{a^2 - \frac{b^2}{4}}$$

Now,

$$\text{Area of } \Delta ABC = \frac{1}{2} \times BC \times AD \qquad \left[\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}\right]$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times b \times \sqrt{a^2 - \frac{b^2}{4}}$$

Important

➥ *In the isosceles triangle, the perpendicular from the vertex angle bisects the base and the vertex angle.*

OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. **A rhombus and an equilateral triangle have equal perimeters. If the perimeter of the rhombus is 60 cm. The altitude of the equilateral triangle is:**

(a) $10\sqrt{3}$ **cm** **(b)** $15\sqrt{3}$ **cm**

(c) $20\sqrt{3}$ **cm** **(d)** $14\sqrt{3}$ **cm**

Ans. *(a)* $10\sqrt{3}$ *cm*

Explanation: Given,

Perimeter of the rhombus = Perimeter of equilateral triangle

$60 = 3 \times$ side of triangle

[Since, the perimeter of an equilateral triangle = $3 \times$ side]

Side of the triangle = $\frac{60}{3} = 20$ cm

Now, the altitude (height) of the equilateral triangle $h = \frac{\sqrt{3}}{2} \times$ side

$$h = \frac{\sqrt{3}}{2} \times 20 = 10\sqrt{3} \text{ cm}$$

2. **Vicky has 3 bamboo sticks of different measures. Now, he wants to increase the measurement to make a triangle by doubling its sides and then reconstructing the triangle. How many times a new perimeter becomes the previous?**

(a) **One time** **(b)** **Four times**

(c) **Two times** **(d)** **Three times**

Ans. *(c) Two times*

Explanation: Let a, b, c be the measurement of bamboo sticks originally.
So, Perimeter of the triangle made up of these sticks, $(P_1) = a + b + c$ .... (i)

If the measurement of bamboo sticks is doubled. Then,
The new sides of the triangle are $2a$, $2b$ and $2c$.
Thus, the new perimeter of the new triangle, (P_2)

$= 2a + 2b + 2c$

$= 2\,[a + b + c]$

$P_2 = 2P_1$ [From eq. (i)]

The new perimeter will be two times the old perimeter.

⚠ **Caution**

- ➥ *The perimeter of a triangle is directly proportional to the sides of the triangle. So, any change inside the triangle will also result in the same change in the perimeter of the triangle.*

Assertion and Reason (A-R)

Direction for the following question: A statement of Assertion (A) is followed by a statement of Reason (R).
Choose the correct option as:

3. **Assertion (A):If the sum of two sides of a triangle is 12 cm and the semi-perimeter of the triangle is 10 cm, then the third side of the triangle is 8 cm.**

   Reason (R):Semi-perimeter of a triangle is $\frac{a+b+c}{2}$.

Ans.(a) *Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).*

Explanation: Let a, b and c be the sides of a triangle.

Let $a + b = 12$ cm

We have, the semi-perimeter of the triangle $= \frac{a+b+c}{2}$

$$10 = \frac{12+c}{2}$$

$$10 \times 2 = 12 + c$$

$$c = 20 - 12$$

$$c = 8 \text{ cm}$$

CASE BASED Questions (CBQs)

[**4** & **5** marks]

Read the following passages and answer the questions that follow:

4. *Ram has shifted from a flat to a bungalow and planted a rose garden in his bungalow's triangular field. The sides of a triangular field are 82 m, 80 m and 18 m. In the garden, each rose bed occupies 20 m^2 area of the field.*

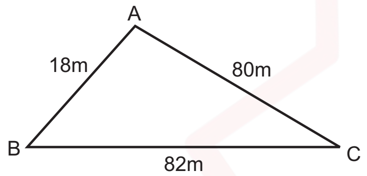
(A)Determine the length of boundary of the triangular field.

(B)Determine the semi-perimeter of the triangular field.

(C)Find the area of the triangular field. If each rose bed occupies 20 m^2 area of the field, then find the number of rose beds in the garden.

Ans. (A)Length of the boundary of the field is the perimeter of the triangular field.

Therefore,



Required length of the boundary

$= AB + AC + BC$

$= 18 \text{ m} + 80 \text{ m} + 82 \text{ m}$

$= 180 \text{ m}$

(B)The semi-perimeter of the triangular field is half of the perimeter of the triangle.

Semi-perimeter (s) $= \frac{\text{Perimeter}}{2}$

$s = \frac{180}{2} \text{ m} = 90 \text{ m}$

(C)Area of the triangular field

$= \sqrt{s(s-a)(s-b)(s-c)}$

Area of ΔABC

$= \sqrt{90(90-18)\,(90-80)\,(90-82)}$

$= \sqrt{9 \times 10 \times 72 \times 10 \times 8}$

$= \sqrt{9 \times 10 \times 9 \times 8 \times 10 \times 8}$

$= 720 \text{ m}^2$

Since, each rose bed occupy = 20 m^2

Therefore, the required number of rose beds

$$= \frac{\text{Total area of triangular field}}{\text{Area of each bed rose}}$$

$$= \frac{720 \text{ m}^2}{20 \text{ m}^2}$$

$$= 36$$

Hence, the number of rose beds = 36

VERY SHORT ANSWER Type Questions (VSA)

[**1** mark]

5. The perimeter of an equilateral triangle is 12 cm. Find the height of the equilateral triangle.

Ans. Since, the sides of an equilateral triangle are congruent. Let a be the side of the equilateral triangle.

Perimeter of the equilateral triangle

$$= 3 \times a$$

$$12 = 3 \times a$$

$$a = 4 \text{ cm}$$

The height of the equilateral triangle

$$= \frac{\sqrt{3}}{2} a$$

$$= \frac{\sqrt{3}}{2} \times 4 = 2\sqrt{3} \text{ cm.}$$

SHORT ANSWER Type-I Questions (SA-I)

[**2** marks]

6. The sides of a triangular field are 51 m, 37 m and 20 m. Find the number of flower beds that can be prepared if each bed is to occupy 6 m^2 of area. [Delhi Gov. QB 2022]

Ans. Sides of a triangular field are 51 cm, 37 cm and 20 cm.

$$\Rightarrow \quad s = \frac{51+37+20}{2}$$

$$= \frac{108}{2} = 54 \text{ m}$$

Area of triangular field

$$= \sqrt{54(54-51)(54-37)(54-20)}$$

$= \sqrt{54 \times 3 \times 17 \times 34}$

$= 306 \text{ m}^2$

Area of each rose bed = 6 m^2

Number of rose beds = $\frac{306}{6}$ = 51

SHORT ANSWER Type-II Questions (SA-II)

[**3** marks]

7. **The sides of a triangular park of a school are 120 m, 100 m and 110 m. The school principal gave a contract to a company to plant grass in the park at the rate of ₹4500 per hectare. How much does the school have to pay to the company? Give your answer to the nearest ₹100.**
[British Council 2022]

Ans. Semi-perimeter = $\frac{120+100+110}{2}$

$= \frac{330}{2}$

$= 165$

Area of triangle

$= \sqrt{165(165-120)(165-100)(165-110)}$

$= \sqrt{165 \times 45 \times 65 \times 55}$

$= 825\sqrt{39} = 825 \times 6.245$

$= 5152.12 \text{ m}^2$ [1 hectare = 10000 sq. m]

$= 0.5152$ hectare

Cost of planting trees = 4500×0.5152

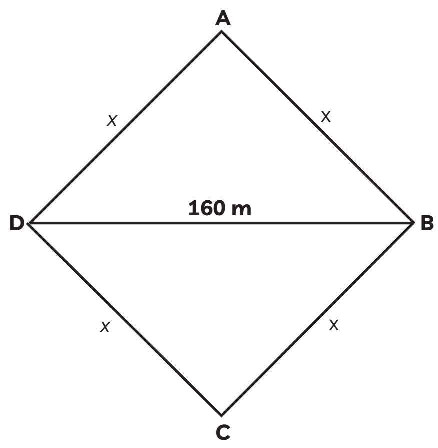
= ₹2318

= ₹2300 (to nearest 100)

LONG ANSWER Type Questions (LA)

[**4** & **5** marks]

8. **Anita has a piece of land which is in the shape of a rhombus. She has two children. One daughter Preetika and one son Naresh to work on the land and produce different types of crops. She divided the land in two equal parts. If the perimeter of the land is 400m and one of the diagonal is 160 m, then how much area will each of them get for their crops?**



[Delhi Gov. QB 2022]

Ans. Let ABCD is the field.

each side be x.

Perimeter of the field = $4x$.

Given perimeter = 400 m

$\Rightarrow \quad 4x = 400$ m

$\Rightarrow \quad x = 100$ m

Length of the diagonal BD = 160 m

In ΔABD

AB = a = 100 m,

AD = b = 100 m and BD = c = 160 m

$$\Rightarrow \quad s = \frac{a+b+c}{2}$$

$$= \frac{100+100+160}{2}$$

$$= 180 \text{ m}$$

$s - a = 180 - 100 = 80$ m

$s - b = 180 - 100 = 80$ m

$s - c = 180 - 160 = 20$ m

$$\text{Area of } \Delta\text{ABD} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{(180 \times 80 \times 80 \times 20)}\ \text{m}$$

$$= \sqrt{9 \times 20 \times 80 \times 80 \times 20}$$

$$= 3 \times 20 \times 80 \text{ m}^2$$

$$= 4800 \text{ m}^2$$

Since, ABCD is a rhombus and diagonal divides it into two congruent triangles.

Therefore, each of them will get an area 4800 m^2.